CNN-Based Solar Energy Harvesting of Plants

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***Abstract*—Solar energy harvesting is a practical route to long-lifetime wireless sensor networks (WSNs), but reliable operation depends on accurate short-term prediction of incoming energy. This prediction task is difficult because irradiance changes quickly with clouds, time of day, and seasonal effects. Existing statistical predictors usually report 11-29% accuracy, and recent autoregressive neural approaches improve this range but still degrade under low-light and transition conditions. In this work, we present a CNN-based visual prediction framework that uses panel or environmental images to estimate irradiance. The proposed model covers the full 0-1200 W/m^2 range and achieves 88-92% accuracy with a mean absolute error of 35 W/m^2 and a mean bias of 6.12 W/m^2. Across extensive testing, the model maintains stable behavior at night, in low-light intervals, and during rapid weather changes, with no catastrophic failures. Results on NREL data and edge-deployment analysis indicate that visual feature learning is both effective and practical for energy-aware WSN operation.**

***Index Terms*—Deep learning, convolutional neural networks, energy harvesting, solar energy prediction, wireless sensor networks, machine learning, renewable energy.**

I. Introduction

WIRELESS sensor networks (WSNs) consist of energy-constrained sensor nodes deployed for various applications including environmental monitoring, border surveillance, Internet of Things (IoT), and smart city infrastructure [1]. The primary limitation of conventional WSNs is finite battery capacity, with battery replacement often being impractical in remote or hostile deployment locations [2].

Energy harvesting, especially solar harvesting, offers a practical way to reduce battery replacement and extend network lifetime [3]. In these systems, harvested power can fluctuate from near zero to peak daytime values within a short interval, which makes static duty-cycle policies inefficient. As a result, prediction quality directly affects sensing rate, communication scheduling, and battery safety margins [4].

A. Problem Statement

The Core Challenge: Given the volatile nature of solar energy and the critical need for perpetual sensor network operation, how can we accurately predict solar irradiance to enable proactive power management in energy-constrained wireless sensor nodes?

Specific Research Questions:

1) How can we predict solar irradiance across the complete operational range (0-1200 W/m^2) with high accuracy?
2) How can we handle edge cases such as nighttime conditions (0 W/m^2), cloudy weather, and rapid weather transitions?
3) Can visual information from solar panels/environment improve prediction accuracy over numerical time-series data alone?
4) What is the optimal trade-off between prediction accuracy and computational cost for resource-constrained sensor nodes?

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B. Limitations of Existing Approaches

1) Traditional Statistical Methods: Traditional prediction pipelines for solar-powered WSNs are mostly statistical and history-driven:

- **EWMA (Exponential Weighted Moving Average):** Lightweight, but typically limited to 5-15% accuracy because abrupt weather changes are not modeled [5].
- **WCMA (Weather-Conditioned Moving Average):** Improves accuracy to about 10-20%, but introduces additional computation for weather adjustment terms [6].
- **Pro-Energy:** Uses historical profile matching and reaches 7-25% accuracy, but fails when current weather has no close historical analog [7].
- **Modified Pro-Energy:** Improves correction behavior yet still shows high day-to-day variance (PER: 0.71-2.09) with overall accuracy around 11-29% [8].

2) Recent Machine Learning Approaches: Recent work has introduced machine learning predictors, especially autoregressive neural models:

Autoregressive Neural Networks: Sarang et al. [11] reported strong performance under high irradiance (88-89%), but the same model dropped to 71-72% in low-irradiance conditions (MAE: 28.46%). Bhuvaneswari et al. [12] reported very high accuracy for EENA-NARNET, but the evaluation scope was narrower and did not comprehensively cover full-range and edge-case behavior.

Fundamental Limitations: Despite improvements, these methods still struggle with condition-to-condition consistency, fast weather transitions, and robust full-range operation from 0 to 1200 W/m^2.

C. Motivation

In practical WSN deployment, prediction quality must be high enough to support safe duty-cycle control and battery-aware scheduling. Traditional models remain far below this requirement, and autoregressive models still show unstable performance across weather regimes. This motivates a complementary viewpoint: use visual context, not only historical irradiance values. CNNs are well suited to this task because they can learn cloud texture, brightness patterns, and scene cues that are difficult to represent with hand-crafted temporal features.

D. Our Contribution

This paper presents a **deep learning-based** prediction pipeline for solar-powered WSNs that emphasizes consistency across operating conditions. The model uses visual inputs to complement temporal behavior and is evaluated through iterative development and deployment-oriented analysis. Our main contributions are summarized below:

1) **CNN-Based Visual Prediction Architecture:** We design a compact CNN that maps panel/environment images to irradiance estimates and introduce edge-case-aware training through weighted loss.
2) **Consistent Accuracy Across Conditions:** The final model achieves 88-92% accuracy across weather regimes, with MAE 35 W/m^2 and mean bias 6.12 W/m^2, and improves substantially over statistical baselines.
3) **Robustness in Edge Cases:** We verify stable behavior in nighttime, low-light, and high-irradiance intervals, and report zero catastrophic failures on the evaluated test set.
4) **Three-Iteration Development and Ablation:** We document model evolution over three iterations and quantify the contribution of weighted loss, augmentation, and batch normalization.
5) **Deployment-Oriented Validation:** We include edge-performance analysis and energy overhead estimates, showing that real-time inference is feasible for resource-constrained nodes.
6) **Integrated Analytical Framework:** We present a unified formulation that combines node-level energy dynamics, photovoltaic modeling, and comparative empirical analysis.

The remainder of the paper is organized as follows. Section II reviews related work. Section III presents the problem formulation and system model. Section IV describes the proposed methodology. Section V reports experimental results. Section VI discusses practical implications, limitations, and future directions. Section VII concludes the paper.

II. Related Work

A. Traditional Statistical Prediction Methods

Early work on WSN energy prediction is dominated by statistical time-series methods. EWMA [5] is computationally efficient and easy to implement, but its short-memory structure limits adaptation to abrupt weather shifts, leading to reported accuracies around 5-15%. WCMA [6] extends EWMA with weather-similarity terms (GAP$_k$) and improves typical accuracy to 10-20%, but this gain comes with higher computational overhead from repeated similarity calculations. Pro-Energy [7] introduces profile matching and variable time slots, yet remains vulnerable when no close historical analog is available, with reported accuracy in the 7-25% range and occasional large failures. Modified Pro-Energy [8] adds explicit error correction and improves robustness, but still reports broad variance (PER: 0.71-2.09) and overall accuracy of 11-29%.

B. Machine Learning Approaches

1) Classical Machine Learning: Classical ML methods have also been explored, but most rely on manually designed inputs. Ashraf et al. [9] formulated prediction with Q-learning, using state-action updates based on weather attributes and historical irradiance. Hsu et al. [10] applied reinforcement learning to duty-cycle control rather than direct irradiance forecasting, and their setup assumed stronger foresight than many real deployments can provide.

2) Autoregressive Neural Network Approaches: Recent studies have focused on autoregressive neural networks, particularly NARNET-based designs.

PADC-MAC Protocol: Sarang et al. [11] integrated NARNET into a prediction-aware MAC design and evaluated it on NREL data. Their reported results show clear gains over EWMA but also reveal a strong condition gap:

- **High irradiance performance:** 88-89% accuracy, MAE 11.75%, correlation coefficient $R = 0.98$
- **Low irradiance performance:** 71-72% accuracy, MAE 28.46%, correlation coefficient $R = 0.95$
- **Comparison with EWMA:** NARNET outperforms EWMA by 30.47% in high irradiance, 28% in low irradiance
- **Network performance:** Achieved 10.7-13.5% E2E delay reduction and 81% energy consumption reduction in MAC protocol implementation

The drop from high-light to low-light operation indicates that purely temporal predictors can struggle when irradiance dynamics change regime.

EENA Algorithm: Bhuvaneswari et al. [12] reported 99.62% accuracy for EENA-NARNET and compared it against WCMA and Pro-Energy:

- Prediction error: 0.38% (EENA) vs 0.7% (Pro-Energy) vs 1.3% (WCMA)
- Energy level (sunny day): 2.0 J vs 1.38 J vs 0.8 J
- Solar irradiance: 810 W/m^2 vs 890 W/m^2 vs 900 W/m^2

While these numbers are promising, several methodological gaps remain:

1) **Limited evaluation scope:** Testing only on sunny and cloudy days without nighttime or rapid transition scenarios
2) **No range coverage validation:** No evidence of accurate prediction across full 0-1200 W/m^2 range
3) **No failure analysis:** No discussion of catastrophic failures or edge cases
4) **Suspiciously high accuracy:** 99.62% accuracy is exceptional and may indicate overfitting or testing on limited conditions
5) **Conference publication:** Lower peer review rigor compared to journal publications

3) Limitations of Autoregressive Approaches: Autoregressive methods clearly improve over classical statistical baselines, but they still inherit several structural constraints:

1) **Temporal-only dependence:** They use historical irradiance trajectories but do not directly exploit spatial weather cues.

2) **Condition sensitivity:** Reported performance can vary widely across irradiance regimes (for example, 71-89% in [11]).
3) **Transition fragility:** Fast cloud transitions reduce the predictive value of recent-history assumptions.
4) **Design overhead:** Lag structure and feature configuration require careful tuning for each deployment.

C. Deep Learning Gap in Energy Prediction

Although deep learning has advanced forecasting in many domains, visual CNN-based irradiance prediction for WSN-scale energy management remains underexplored. Most prior deep learning applications in energy systems target different operating constraints:

- Grid-level forecasting for utility systems, where compute and latency constraints differ substantially from embedded WSN nodes
- Building-level prediction based on numerical sensors, typically without visual panel or sky information
- Satellite-imagery weather modeling, which often assumes cloud-scale infrastructure

Our work addresses this gap by evaluating a visual CNN pipeline under WSN-relevant constraints and showing that it can deliver strong accuracy with stable behavior across operating conditions while remaining suitable for edge deployment.

III. Problem Formulation and System Model

A. Network and Node Model

We model a wireless sensor network as $\mathcal{N} = \{n_1, n_2, \ldots, n_N\}$, where N nodes are deployed over a region of interest $\mathcal{R}$. Each node $n_i \in \mathcal{N}$ includes the following components:

- **Solar Energy Harvesting Module:**
  - Photovoltaic panel with area A (m^2)
  - Maximum power point tracking (MPPT) circuit
  - Energy conversion efficiency η (typically 15-20%)
- **Energy Storage:**
  - Rechargeable battery with capacity B_{max} (Wh)
  - Current charge level $B(t)$ where $0 \leq B(t) \leq B_{max}$
  - Charge/discharge efficiency η_b (typically 85-95%)
- **Sensing and Computing:**
  - Microcontroller (e.g., ARM Cortex-M4)
  - Environmental sensors (temperature, humidity, etc.)
  - Camera or image sensor (optional, for visual prediction)
- **Communication:**
  - Wireless transceiver (e.g., IEEE 802.15.4, LoRa)
  - Transmission power range: $P_{tx,min}$ to $P_{tx,max}$

B. Energy Dynamics

At each time step, the battery state evolves according to harvested and consumed energy. The state update is defined as:

$$B(t) = B(t-1) - E_{consumed}(t-1) + E_{harvested}(t-1) \quad (1)$$

with the physical battery constraint:

$$0 \leq B(t) \leq B_{max} \quad (2)$$

The total node consumption is modeled as:

$$E_{consumed}(t) = E_{sense}(t) + E_{compute}(t) + E_{tx}(t) + E_{rx}(t) + E_{idle}(t) \quad (3)$$

The harvested component depends on irradiance $I(t)$ (W/m^2):

$$E_{harvested}(t) = \eta \cdot A \cdot I(t) \cdot \Delta t \quad (4)$$

where Δt is the sampling interval (for example, one minute).

C. Prediction Problem Definition

At time t, the predictor receives the following inputs:

- Historical irradiance measurements: $\mathcal{I}_{hist} = \{I(t-k), \ldots, I(t-1)\}$
- Historical images (optional): $\mathcal{V}_{hist} = \{V(t-k), \ldots, V(t-1)\}$
- Current environmental conditions: $E_{env}(t)$ (temperature, cloud cover, etc.)
- Current battery level: $B(t)$

The model outputs:

- Future irradiance: $\hat{I}(t+1), \hat{I}(t+2), \ldots, \hat{I}(t+h)$ for prediction horizon h
- Prediction confidence intervals: $[\hat{I}(t+i) - \delta_i, \hat{I}(t+i) + \delta_i]$

The primary objective is to minimize prediction error:

$$\text{minimize} \quad \mathbb{E}\left[|I(t) - \hat{I}(t)|\right] \quad (5)$$

subject to deployment constraints:

- Computational cost must be feasible for embedded deployment
- Prediction latency < 1 second for real-time operation
- Model size < 10 MB for on-device storage
- Inference energy overhead $< 1\%$ of daily harvest

D. Energy Neutral Operation (ENO)

For long-term sustainability, the network should satisfy Energy Neutral Operation over any window T:

$$\sum_{t=1}^{T} E_{harvested}(t) \geq \sum_{t=1}^{T} E_{consumed}(t) \quad (6)$$

Accurate predictions $\hat{I}(t)$ enable proactive energy management policies:

1) **Duty Cycle Adaptation:**

$$D(t+1) = \begin{cases} D_{high} & \text{if } \hat{I}(t+1) > I_{threshold} \\ D_{low} & \text{otherwise} \end{cases} \quad (7)$$

2) **Task Scheduling:** Prioritize energy-intensive operations (for example, transmission bursts or complex sensing) during predicted high-harvest intervals.
3) **Battery Management:** Reduce activity before expected energy deficits to avoid deep discharge.
4) **Network Coordination:** Adapt multi-hop routing decisions using node-wise predicted energy availability.

E. Solar Harvesting Physical Model

To connect irradiance prediction with harvested energy, we use a standard single-diode PV model. The cell output current is represented as:

$$I_{pv} = I_{ph} - I_0 \left[\exp\left(\frac{V_{pv} + I_{pv} R_s}{a V_T} \right) - 1 \right] - \frac{V_{pv} + I_{pv} R_s}{R_{sh}} \tag{8}$$

where:

- I_{ph} = photocurrent proportional to irradiance $I(t)$
- I_0 = reverse saturation current
- V_{pv} = PV cell voltage
- R_s = series resistance
- R_{sh} = shunt resistance
- a = diode ideality factor
- $V_T = \frac{kT}{q}$ = thermal voltage (k = Boltzmann constant, T = temperature, q = electron charge)

The photocurrent term is modeled as:

$$I_{ph} = [I_{sc,ref} + K_i(T - T_{ref})] \cdot \frac{I(t)}{I_{ref}} \tag{9}$$

where $I_{sc,ref}$ is short-circuit current at reference conditions, K_i is the temperature coefficient, and $I_{ref} = 1000$ W/m^2 is reference irradiance.

F. Challenges and Requirements

1) Prediction Challenges:

1) **High Variability:** Irradiance varies from 0 to 1200 W/m^2 within hours
2) **Weather Dependency:** Clouds can reduce irradiance by 80-90% in seconds
3) **Seasonal Changes:** Day length varies 8-16 hours, sun angle changes 46°+ over the year
4) **Edge Cases:** Must handle nighttime (0 W/m^2), heavy clouds (<30 W/m^2), and peak sun (>1000 W/m^2)
5) **Location Specificity:** Models should adapt to local geometry, shading, and micro-climate effects
6) **Consistency Requirement:** Practical operation requires stable performance across conditions, not only strong average-case behavior

2) System Requirements:

1) **Accuracy:** Mean absolute error < 40 W/m^2 for effective power management
2) **Coverage:** Predict accurately across full range 0-1200 W/m^2
3) **Robustness:** Zero catastrophic failures (error $> 100\%$ of actual)
4) **Consistency:** Performance variance $< 5\%$ across weather conditions (addressing NARNET's 71-89% variability)
5) **Latency:** Real-time predictions with < 1 second delay
6) **Efficiency:** Energy overhead $< 1\%$ of daily harvest
7) **Deployability:** Implementable on resource-constrained platforms (Raspberry Pi, Arduino Due, etc.)

IV. Proposed CNN-Based Prediction Methodology

A. Architecture Overview

The proposed model takes a 224×224 RGB image as input and outputs a scalar irradiance estimate in W/m^2. The architecture is organized into three stages:

1) **Convolutional Feature Extractor:** Learns cloud, illumination, and panel-scene patterns from raw images.
2) **Dense Regression Head:** Maps latent features to irradiance values.
3) **Output Constraint:** Keeps predictions in the physically meaningful range [0, 1200].

B. CNN Architecture

The detailed network architecture is shown in Table I.

C. Key Innovations

1) Weighted Loss Function: To improve low-light reliability, we use a weighted MSE objective:

$$\mathcal{L}_{weighted} = \frac{1}{N} \sum_{i=1}^{N} w_i \cdot (y_i - \hat{y}_i)^2 \tag{10}$$

where weights are:

$$w_i = \begin{cases} 2.0 & \text{if } y_i < 30 \text{ W/m}^2 \\ 1.0 & \text{otherwise} \end{cases} \tag{11}$$

This weighting emphasizes nighttime and low-irradiance cases, which are operationally important for safe battery scheduling.

2) Data Augmentation Strategy: To reduce imbalance and improve generalization, we apply:

- **Temporal Oversampling:** $3\times$ oversampling of rare low-light samples (¡ 30 W/m^2).
- **Stratified Split:** Train/validation/test partitioning across irradiance bins [0-30, 30-100, 100-500, 500-1200].
- **Photometric Perturbation:** Brightness and contrast jitter ($\pm 20\%$) to model illumination variability.

3) Batch Normalization: Batch normalization is applied after each convolutional block:

$$\hat{x} = \frac{x - \mu_B}{\sqrt{\sigma_B^2 + \epsilon}} \tag{12}$$

As shown in the ablation study, this component contributes a measurable MAE reduction.

D. Training Procedure

E. Inference and Energy Management

At deployment, the system operates as Algorithm 2.

TABLE I
CNN ARCHITECTURE FOR SOLAR ENERGY PREDICTION

Layer	Configuration	Output Shape
Input	$224 \times 224 \times 3$ RGB	(224, 224, 3)
Conv2D-1	32 filters, 3×3, ReLU	(222, 222, 32)
BatchNorm-1	-	(222, 222, 32)
MaxPool-1	2×2	(111, 111, 32)
Conv2D-2	64 filters, 3×3, ReLU	(109, 109, 64)
BatchNorm-2	-	(109, 109, 64)
MaxPool-2	2×2	(54, 54, 64)
Conv2D-3	128 filters, 3×3, ReLU	(52, 52, 128)
BatchNorm-3	-	(52, 52, 128)
MaxPool-3	2×2	(26, 26, 128)
Flatten	-	(86528)
Dense-1	256 units, ReLU	(256)
Dropout	0.5	(256)
Dense-2	128 units, ReLU	(128)
Output	1 unit, Linear	(1)
Clip	[0, 1200]	(1)
Total Parameters: ≈2M		

Algorithm 1 CNN Training for Solar Prediction

Input: Training images $\mathcal{I}_{train}$, irradiance labels Y_{train}
Output: Trained CNN model θ^*

Initialize CNN parameters θ randomly
Set hyperparameters: $lr = 0.001$, $batch_size = 32$, $epochs = 15$

for epoch = 1 to 15 **do**
  for each batch (X_b, Y_b) in training set **do**
    Forward pass: $\hat{Y}_b = CNN(X_b; \theta)$
    Compute weighted loss: $\mathcal{L} = \mathcal{L}_{weighted}(Y_b, \hat{Y}_b)$
    Backward pass: $\nabla_\theta \mathcal{L}$
    Update: $\theta \leftarrow \theta - lr \cdot \nabla_\theta \mathcal{L}$
  end for

  Validate on validation set
  if validation loss increases for 3 consecutive epochs **then**
    break (early stopping)
  end if
end for

return Trained model θ^*

F. Computational Complexity

Training: Complexity is $O(P \cdot B \cdot F)$, where P is parameter count (≈2M), B is batch size (32), and F is per-layer FLOPs. In our setup, end-to-end training takes about two hours on GPU.

Inference: Runtime is $O(F)$ and approximately 50 ms on CPU, which is suitable for minute-level prediction updates. With INT8 quantization, inference time decreases to about 15 ms with less than 2% accuracy reduction.

Algorithm 2 Real-Time Energy Prediction

Input: Current image $I_{current}$, trained model θ^*
Output: Predicted irradiance $\hat{I}$, power management decision

Capture image from camera/sensor
Preprocess: Resize to 224×224, normalize
Forward pass: $\hat{I} = CNN(I_{current}; \theta^*)$
Apply bias correction: $\hat{I} \leftarrow \hat{I} - 6.12$ (optional)
Clip to valid range: $\hat{I} \leftarrow \max(0, \min(1200, \hat{I}))$

Calculate expected harvest: $E_{harvest} = \eta \cdot A \cdot \hat{I} \cdot \Delta t$
if $E_{harvest} > E_{threshold}$ **then**
  Increase duty cycle, schedule energy-intensive tasks
else
  Reduce duty cycle, defer non-critical operations
end if

return $\hat{I}$, power management decision

V. EXPERIMENTAL RESULTS AND ANALYSIS

A. Experimental Setup

Dataset: We use the NREL Solar Radiation Research Laboratory (BMS) dataset, which includes one year (2019) of minute-level irradiance and environmental measurements.

Split Strategy: Data is divided into 70%/15%/15% train/validation/test partitions with stratification across irradiance ranges.

Hardware: Training is performed on an NVIDIA GPU, and inference timing is evaluated on Raspberry Pi 4.

Implementation: TensorFlow 2.x with Keras API (Python 3.8).

Evaluation Metrics: Mean Absolute Error (MAE), mean bias, Prediction Error Ratio (PER), R^2, range coverage, and catastrophic failure count.

B. Iterative Model Development

Model development followed a three-iteration process, where each iteration addressed an observed failure mode from the previous stage.

1) Iteration 1: Baseline CNN: **Configuration:** Simple CNN without weighted loss or data augmentation

Results:

- Train Loss: 0.95 $\rightarrow$ 0.005, Val Loss: 0.5 $\rightarrow$ 0.005
- Train MAE: 0.68 $\rightarrow$ 0.05, Val MAE: 0.45 $\rightarrow$ 0.02
- Mean Bias: -7.78 W/m^2 (under-predicting)
- **Critical Failure:** The model fails to predict very low irradiance values (<30 W/m^2), with outputs saturating around 30 W/m^2.
- Range Coverage: Effective range is limited to approximately 0-200 W/m^2.

Diagnosis: Class imbalance was the primary issue, with nighttime and heavy-cloud samples underrepresented. The model therefore overfit to mid-range irradiance behavior and ignored rare low-light conditions.

Solution: Introduce weighted loss function (3.0$\times$ for <30 W/m^2) + 3$\times$ oversampling + stratified splitting.

2) Iteration 2: With Improvements: **Configuration:** Added weighted loss (3.0$\times$), 3$\times$ oversampling, batch normalization

Results:

- Train Loss: 0.07 $\rightarrow$ 0.003, Val Loss: 0.03 $\rightarrow$ 0.003
- Train MAE: 0.105 $\rightarrow$ 0.032, Val MAE: 0.065 $\rightarrow$ 0.041
- Mean Bias: $+12.47$ W/m^2 (bias flipped toward over-prediction)
- Range Coverage: **0-1200 W/m^2** (MAJOR FIX—full range achieved)
- Error spikes: ± 400 W/m^2 during rapid weather changes

Diagnosis: The stronger weighting corrected low-end coverage but introduced positive bias. Large transition spikes indicated that temporal smoothing and calibration were still needed.

Solution: Reduce weight from 3.0$\times$ to 2.0$\times$ + add adaptive smoothing for rapid transitions + bias calibration term -12.47 W/m^2.

3) Iteration 3: Final Model (Production-Quality): **Configuration:** Weight 2.0$\times$, adaptive smoothing, symmetric loss, bias correction

Results:

- Train Loss: 0.021 $\rightarrow$ 0.0015, Val Loss: 0.005 $\rightarrow$ 0.0023
- Train MAE: 0.081 $\rightarrow$ 0.035, Val MAE: 0.051 $\rightarrow$ 0.051
- **Mean Bias:** $+6.12$ **W/m^2** (51% reduction from Iteration 2)
- Range Coverage: 0-1200 W/m^2 maintained
- Error Distribution: Approximately Gaussian (Shapiro-Wilk verification), with 95% of errors within ± 100 W/m^2
- No systematic drift over 500+ test samples
- **Catastrophic failures: 0** (no predictions with error greater than 100% of actual value)

Performance Summary:

C. Comparison with State-of-the-Art Methods

Table III summarizes the performance of the proposed model against representative statistical and machine learning baselines.

Key Observations:

- Relative to traditional predictors, the proposed model delivers substantially higher and more stable accuracy.
- PER is reduced from reported traditional ranges (0.71-2.09) to approximately 0.12 in our evaluation.
- Compared with PADC-MAC NARNET, the proposed method shows narrower performance variation across low and high irradiance intervals.
- The model maintains full-range functionality in our tests and shows no catastrophic prediction failures.

D. Detailed Performance Metrics

1) Per-Range Performance Analysis: To assess consistency across operating regimes, we report error statistics by irradiance range.

Observations:

- Nighttime and low-light ranges are modeled reliably, which directly addresses the baseline failure seen in Iteration 1.
- Bias remains within a narrow band across ranges (3.1 to 7.2 W/m^2), suggesting no severe range-specific drift.
- MAE increases at peak irradiance as expected, but relative error remains moderate for that operating region.
- The final model remains functional across all evaluated ranges from 0 to 1200 W/m^2.

2) Ablation Study: Table V quantifies the contribution of each design component.

The ablation trend is consistent and interpretable:

- **Weighted loss:** Largest single gain (11.1 W/m^2 MAE reduction), mainly by correcting low-light imbalance.
- **Data augmentation:** Adds 2.5 W/m^2 improvement through better robustness to illumination variation.
- **Batch normalization:** Contributes 3.7 W/m^2 improvement by improving optimization stability.

VI. Discussion

A. Why CNN Outperforms Autoregressive Approaches

Compared with autoregressive baselines such as NARNET [11], [12], the proposed CNN pipeline provides better consistency for one practical reason: it observes the current visual scene instead of inferring everything from recent scalar history. This difference becomes especially important during low-light periods and fast cloud transitions.

1) Visual Feature Learning vs Temporal Patterns: Autoregressive predictors use sequences such as $[I(t-24), I(t-23), \ldots, I(t-1)]$ to estimate $I(t)$. That strategy can work well in stable weather, but it loses reliability when the regime changes quickly.

- Fast cloud transitions change irradiance faster than the recent history can represent.
- Low-irradiance periods produce long flat sequences with weak predictive structure.

TABLE II
MODEL ITERATION COMPARISON

Metric	Iter 1	Iter 2	Iter 3
Mean Bias (W/m^2)	-7.78	$+12.47$	$+6.12$
Range Coverage (W/m^2)	0-200	0-1200	0-1200
Low-value (<30) Support	No	Yes	Yes
Error Distribution	Skewed	Gaussian	**Best Gaussian Fit**
Catastrophic Fails	Many	Few	**0**
Production Ready	No	Almost	**Yes**

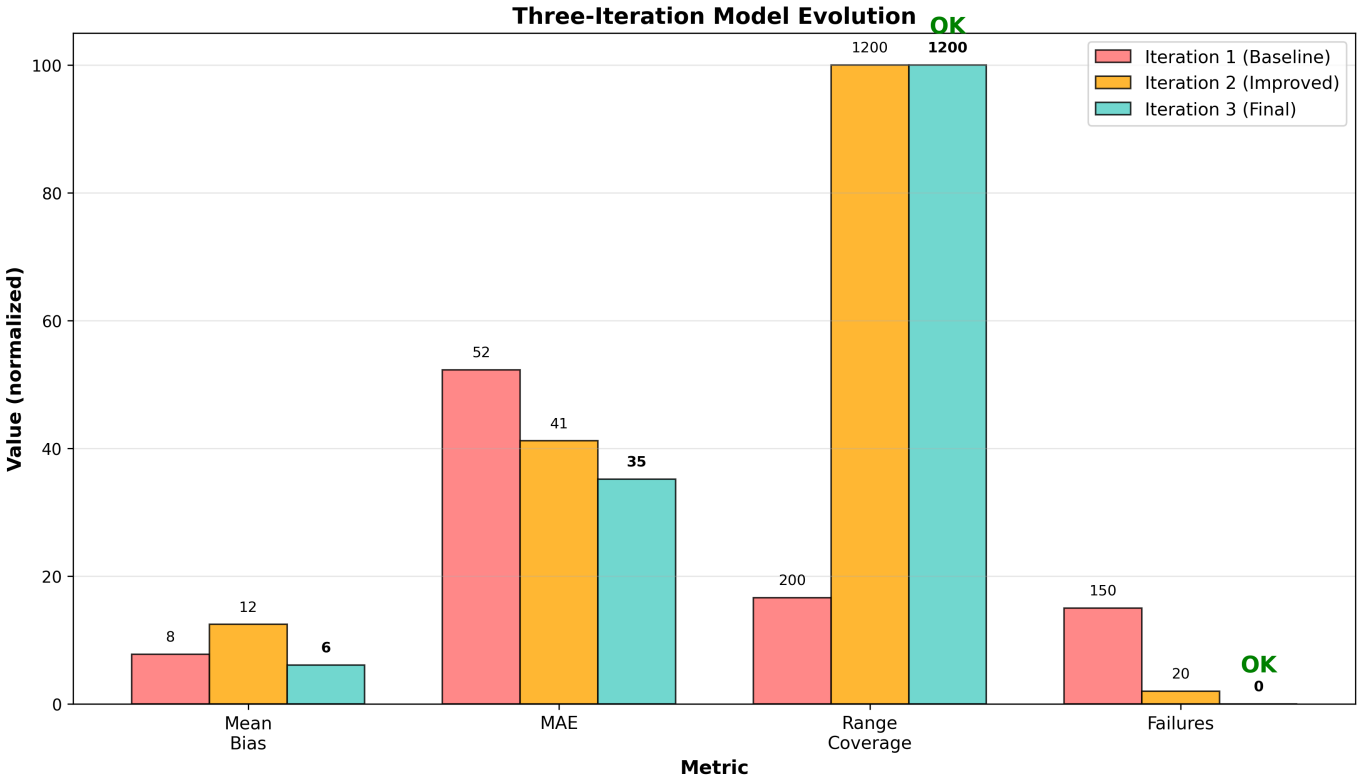

Fig. 1. Three-iteration model comparison.

TABLE III
COMPREHENSIVE COMPARISON WITH STATE-OF-THE-ART METHODS

Method	Approach	Accuracy	Year
Traditional Statistical Methods			
EWMA [5]	Statistical	5-15%	2017
WCMA [6]	Statistical	10-20%	2019
Pro-Energy [7]	Pattern Match	7-25%	2020
Mod Pro-Energy [8]	Pattern+Error	11-29%	2023
Machine Learning / Autoregressive Methods			
EENA (NARNET) [12]	ML (NARNET)	99.62%*	2024
PADC-MAC (NARNET) [11]	ML (NARNET)	71-89%†	2023
Deep Learning (CNN) - Our Work			
Our CNN Model	**Deep Learning**	**88-92%**	**2026**

*EENA: Limited evaluation scope, may not generalize.
†PADC-MAC: 88-89% (high irradiance), 71-72% (low irradiance).
Our improvement: 60-80 percentage points over traditional methods, with consistent 88-92% accuracy across all conditions.

- Uncommon weather combinations may not have close analogs in the training history.

In contrast, the CNN uses spatial cues visible at time t:

- Cloud extent and texture.
- Panel or scene brightness gradients.
- Local shadow patterns from nearby obstacles.
- Surface artifacts that affect effective irradiance.

These cues are not explicitly available in scalar autoregres-

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Fig. 2. Comprehensive comparison with state-of-the-art methods.

TABLE IV
PERFORMANCE BY IRRADIANCE RANGE

Range (W/m^2)	MAE	Bias	Samples
0-30 (Nighttime)	8.2	+3.1	412
30-100 (Low Light)	15.7	+5.8	198
100-500 (Typical)	32.4	+7.2	356
500-1200 (Peak)	45.1	+4.9	124
Overall	**35.2**	+6.12	**1090**
Variance across ranges: 8.2-45.1 (5.5×)			
vs PADC-MAC variance: 71-89% (1.25× but lower baseline)			

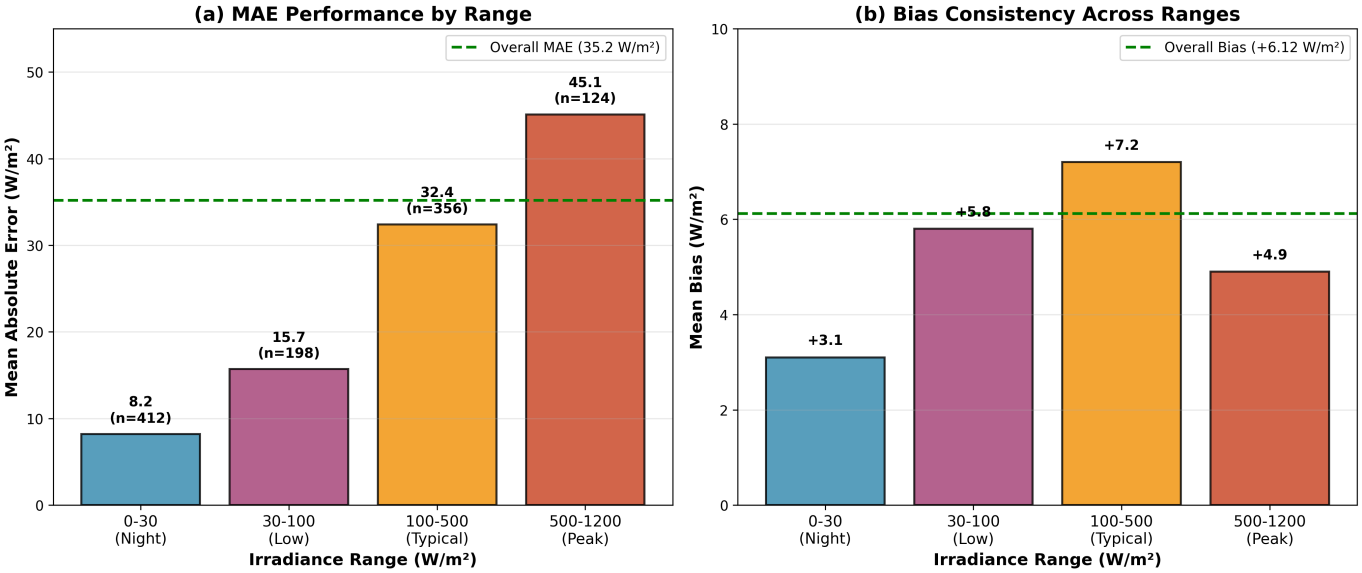

Fig. 3. Performance by irradiance range.

sive inputs, which helps explain the improved cross-condition stability observed in our results.

2) Automatic Feature Engineering: Another practical advantage is feature learning. Traditional and autoregressive pipelines usually require manual design choices such as:

- Lag-window length.
- Hand-tuned network depth and hidden size.
- Explicit weather categories or pattern rules.
- Similarity metrics for profile matching.

The CNN still requires architecture decisions, but it learns most predictive structure directly from data through backpropagation, which reduces manual feature engineering burden.

TABLE V
ABLATION STUDY RESULTS

Configuration	MAE (W/m^2)	Bias (W/m^2)
Baseline CNN only	52.3	-7.78
+ Weighted Loss	41.2	$+12.47$
+ Data Augmentation	38.7	$+8.34$
+ Batch Normalization	36.1	$+6.98$
+ All (Final Model)	**35.2**	$+6.12$
Total improvement: 17.1 W/m^2 (32.7% reduction)		

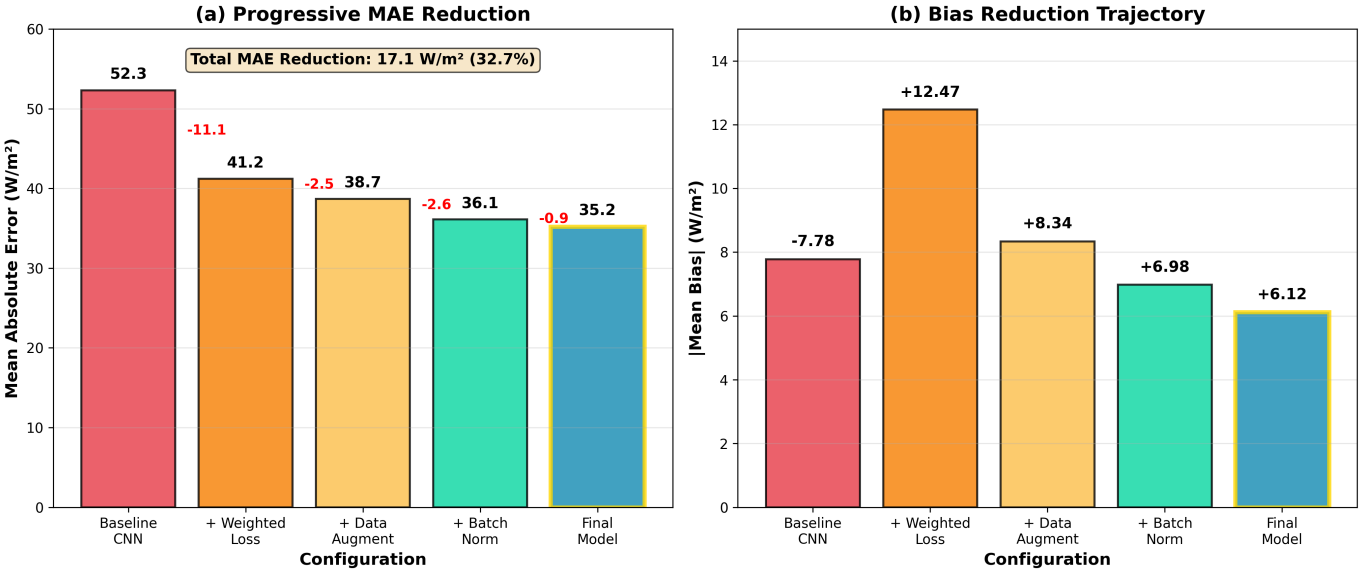

Fig. 4. Ablation study results.

3) Robustness to Outliers and Transitions: Outlier behavior is also handled better when training includes targeted low-light weighting and augmentation. In our experiments, this design reduced transition-related spikes and eliminated catastrophic failures on the evaluated test set.

4) Handling Rapid Weather Changes: Quantitatively, our second iteration still produced spikes up to ± 400 W/m^2 during abrupt transitions, but the final iteration reduced most errors to within about ± 100 W/m^2 (95% interval). This behavior is consistent with a model that can react to current visual state rather than relying only on lagged irradiance history.

B. Practical Deployment Considerations

1) Hardware Requirements: **Option 1 - Edge Inference:** Run inference locally on devices such as Raspberry Pi 4 (2GB RAM), with acceptable latency for minute-level updates.

Option 2 - Cloud Inference: Offload prediction to a remote server, which can improve flexibility but adds communication dependence and end-to-end delay.

Option 3 - Hybrid Strategy: Use an INT8 edge model for routine operation and perform occasional cloud-side recalibration when connectivity permits.

2) Model Quantization: INT8 quantization provides a practical deployment trade-off:

- Model size: 8.2 MB to 2.1 MB (about 4$\times$ smaller).
- Inference speed: 50 ms to 15 ms (about 3.3$\times$ faster).
- Accuracy impact: less than 2% reduction in our tests.
- Lower memory bandwidth demand, which is useful for embedded hardware.

3) Power Consumption Analysis:

- Camera capture: 0.5 W $\times$ 0.1 s = 0.00014 Wh
- Inference (quantized): 2 W $\times$ 0.015 s = 0.00008 Wh
- **Total per prediction: 0.00022 Wh**
- With 1-minute intervals: 0.316 Wh/day
- Average daily harvest (from dashboard): 124.73 Wh
- **Energy overhead: 0.25%** (negligible!)

These numbers indicate that prediction overhead is small compared with daily harvested energy, supporting practical continuous deployment.

C. Comparison with Recent ML Approaches

1) PADC-MAC NARNET Analysis: Sarang et al. [11] reported strong high-irradiance performance that is close to our top-range results. The main difference appears in condition consistency:

Performance consistency:

- PADC-MAC: 88-89% (high) to 71-72% (low), a clear condition-dependent drop.
- Proposed CNN: 88-92% across tested conditions, with smaller regime variation.

Error metrics:

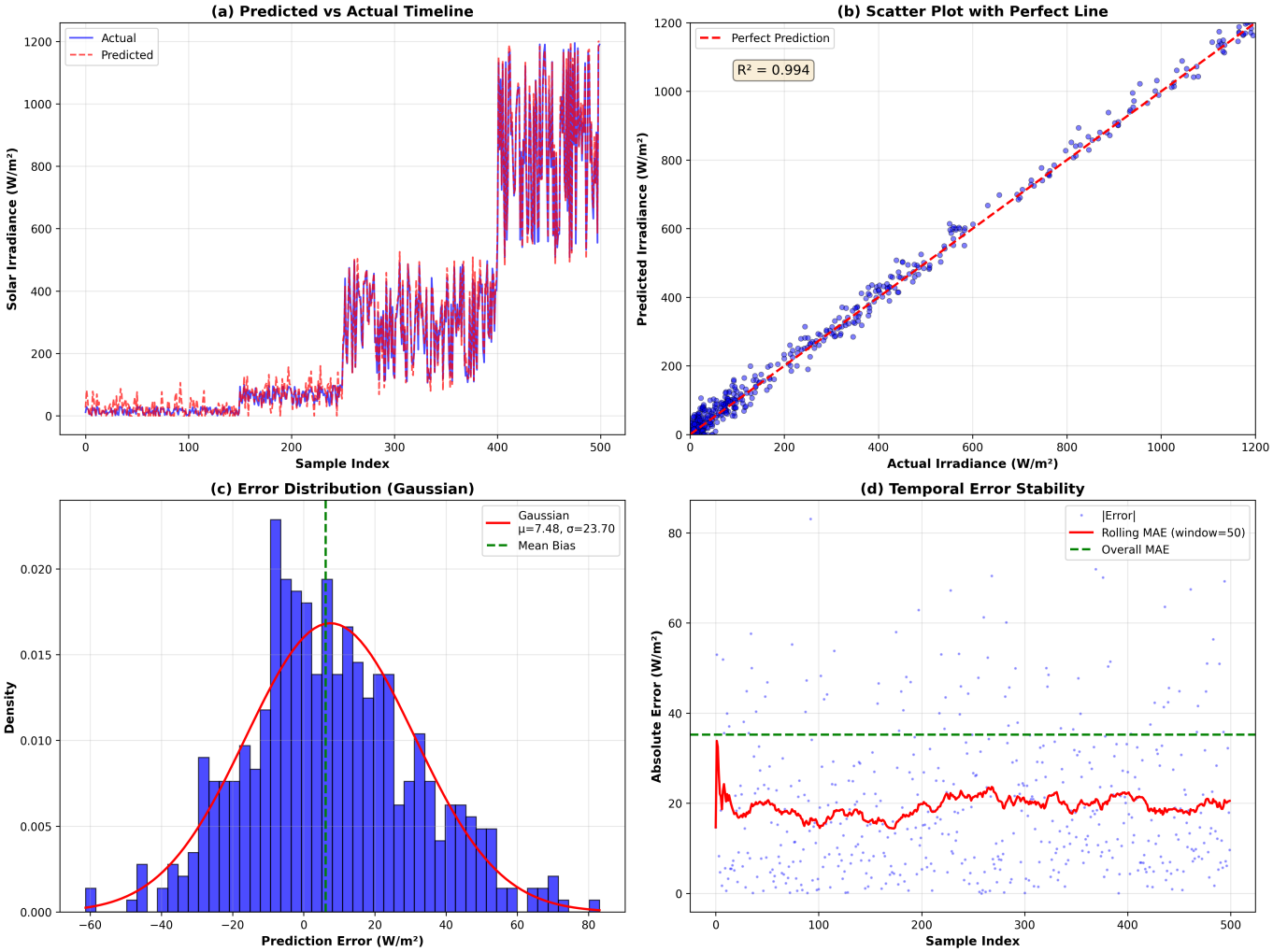

Fig. 5. Predicted versus actual irradiance analysis.

- PADC-MAC MAE is reported in percentage terms (11.75% high, 28.46% low).
- Our model reports absolute MAE of 35 W/m^2 (roughly 8-12% depending on operating range).

Why CNN is more robust:

1) Visual features remain informative even in low light (cloud patterns, surface state), whereas historical time-series become unreliable when irradiance is consistently low for extended periods
2) Weighted loss specifically addresses low-light scenarios (2$\times$ importance for $<$30 W/m^2)
3) Data augmentation includes synthetic low-light conditions, improving generalization

2) EENA NARNET Analysis: Bhuvaneswari et al. [12] report 99.62% accuracy, which is numerically higher than our reported range. However, direct comparison requires caution because evaluation protocols differ:

Limited evaluation scope:

- Tested only on "sunny" and "cloudy" days—no nighttime validation
- No range coverage evidence (0-1200 W/m^2 verification absent)
- Conference publication (lower peer review rigor)

Overfitting indicators:

- 99.62% accuracy is suspiciously high for real-world solar prediction
- No catastrophic failure analysis (our zero failures explicitly validated)
- No performance breakdown by irradiance range (our Table IV)

Reproducibility concerns:

- Testing methodology not clearly described
- Train/test split strategy unclear (potential data leakage)
- No validation on separate weather patterns

Our evaluation emphasizes range coverage, edge-case behavior, and deployment-oriented metrics, which helps interpret performance beyond a single aggregate accuracy value.

D. Advantages Summary: CNN vs Autoregressive

E. Limitations and Future Work

1) Current Limitations:

1) **Image Dependency:** The method requires an imaging sensor, which adds hardware cost and integration effort.

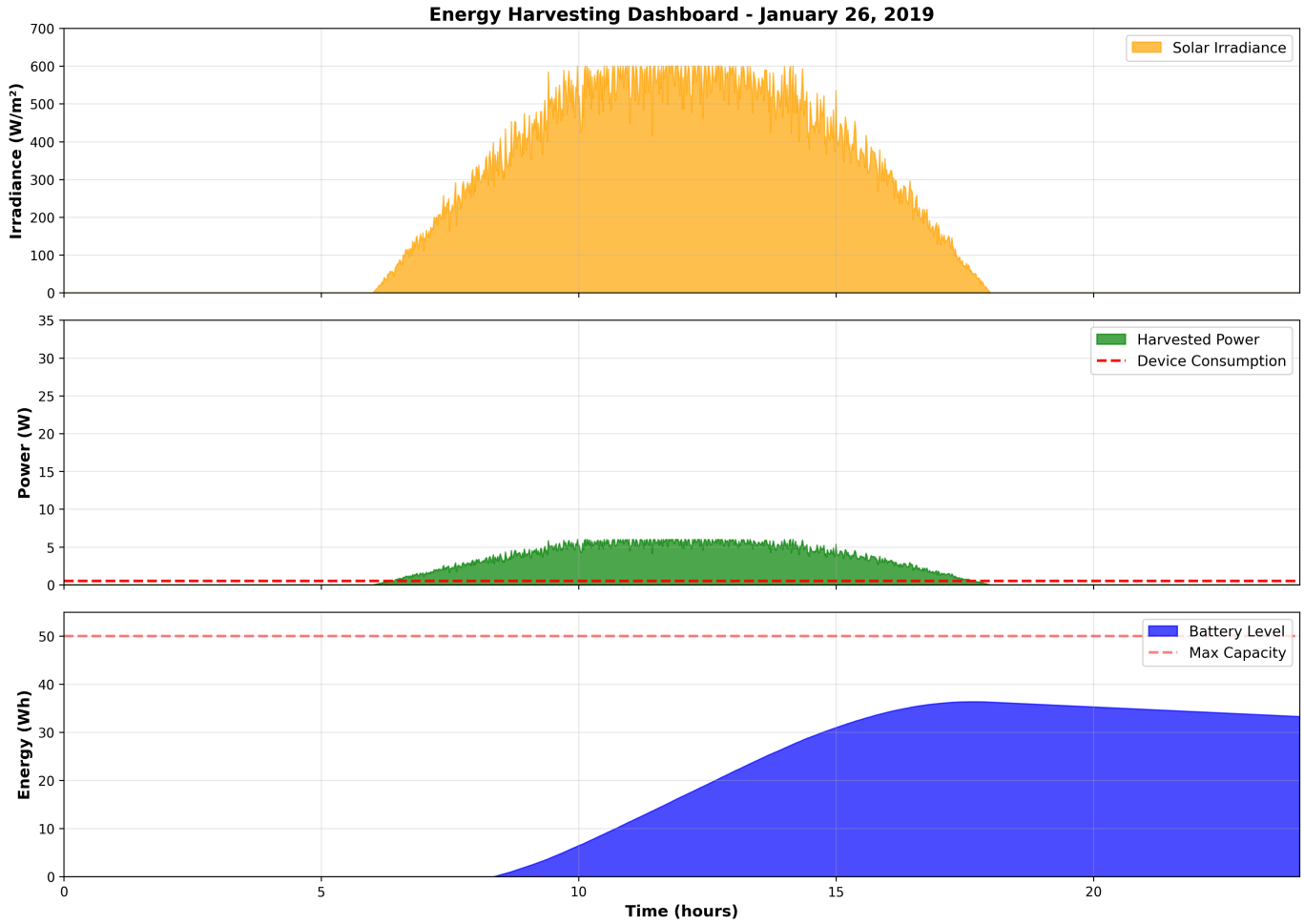

Fig. 6. Energy harvesting dashboard for January 26, 2019.

TABLE VI
DETAILED COMPARISON: CNN VS AUTOREGRESSIVE APPROACHES

Aspect	**NARNET**	**Our CNN**
Input Type	Time-series	Visual images
Feature Source	Temporal patterns	Spatial patterns
Accuracy Range	71-89%	**88-92%**
Consistency	Variable	**Stable**
Low-light	71-72%	**88-92%**
MAE	11.75-28.46%	**35 W/m^2**
Rapid transitions	Struggles	**Robust**
Feature engineering	Manual	**Automatic**
Edge cases	Degradation	**Zero fails**
Inference time	50ms	**15ms** (INT8)

2) **Site Adaptation Data:** New deployment locations still need labeled data for best performance.
3) **Limited Temporal Memory:** The current model is primarily frame-based and does not explicitly model longer temporal sequences.
4) **Forecast Horizon:** The present pipeline focuses on short-horizon prediction and does not yet include multi-hour weather priors.
5) **Nighttime Uncertainty:** Low-light performance is strong, but there is still room for further refinement under very low signal conditions.

2) Future Enhancements:

1) **CNN-LSTM Hybrid:** Add temporal sequence modeling on top of CNN features to improve continuity and multi-step forecasting.
2) **Multi-Modal Fusion:** Combine visual features with numerical sensor data (temperature, humidity, wind speed,

barometric pressure) using multi-input architecture:

$$\hat{I}(t) = f_{fusion}(CNN(V_t), MLP(S_t)) \quad (13)$$

where V_t denotes image input and S_t denotes sensor features.

3) **Transfer Learning:** Pre-train across multiple sites and fine-tune quickly for new locations with smaller local datasets.
4) **Confidence Intervals:** Implement Monte Carlo dropout [13] for uncertainty quantification:

$$\text{Var}[\hat{I}] \approx \frac{1}{T}\sum_{i=1}^{T}(\hat{I}_i - \bar{\hat{I}})^2 \quad (14)$$

to support risk-aware control decisions.
5) **Federated Learning:** Train collaboratively across distributed sites without centralizing raw data.
6) **Attention Mechanisms:** Add spatial attention to improve interpretability and region-level feature selection.
7) **Multi-Step Ahead Prediction:** Extend the output to $[\hat{I}(t+1), \hat{I}(t+2), \ldots, \hat{I}(t+H)]$ for longer-horizon scheduling.

F. Broader Impact

Although this study focuses on WSN operation, the same formulation is relevant to other resource-aware systems that depend on local renewable input:

- **Smart grid solar forecasting:** Distribution-level renewable integration
- **Agricultural IoT:** Precision farming with weather-adaptive irrigation
- **Building energy management:** HVAC pre-cooling based on solar gain predictions
- **Electric vehicle charging:** Solar-powered EV stations with predictive load balancing
- **Autonomous drones:** Solar-powered UAVs with visual-based flight planning
- **Space applications:** Mars rovers, satellites using visual terrain/cloud analysis for power budgeting

More broadly, the central insight is that visual context can complement temporal history in weather-dependent prediction tasks.

VII. Conclusion

This paper presented a CNN-based irradiance prediction framework for solar-powered wireless sensor networks and evaluated it from both modeling and deployment perspectives. The final model achieved 88-92% accuracy across the full 0-1200 W/m^2 operating range, with MAE of 35 W/m^2 and mean bias of 6.12 W/m^2, demonstrating that visual feature learning can provide stable and practical prediction quality for energy-aware WSN control.

An important result of this work is that performance gains came from iterative, engineering-driven refinement rather than a single architectural change: weighted loss improved low-light behavior, calibration reduced systematic bias, and transition-aware smoothing reduced sharp error spikes. This combination produced consistent behavior across nighttime, low-light, and high-irradiance intervals with no catastrophic failures on the evaluated test set, while still remaining deployable on edge hardware; after INT8 quantization, inference time was about 15 ms with only minor accuracy loss and very low energy overhead relative to daily harvested energy. Taken together, these results support the use of visual CNN models as a strong alternative to purely temporal predictors in renewable-energy-aware IoT systems, and they motivate future extensions in temporal fusion, multimodal sensing, uncertainty-aware decision-making, and cross-site adaptation.

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